

Appendix S1

“Successful application of ancient DNA extraction and library construction protocols to museum wet collection specimens”

Authors: Nicolas Straube, Mariana L. Lyra, Johanna L. A. Paijmans, Michaela Preick, Nikolas Basler, Johannes Penner, Mark-Oliver Rödel, Michael V. Westbury, Célio F. B. Haddad, Axel Barlow and Michael Hofreiter.

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Supplementary Material 1

ARCHIVAL DNA EXTRACTION FROM WET COLLECTION SPECIMEN TISSUE SAMPLES

1 ESSENTIAL READING

This protocol is based on and modified from the references listed below. If using this protocol, please read and cite:

- Basler, N., Xenikoudakis, G., Westbury, M. V., Song, L., Sheng, G., & Barlow, A. (2017). Reduction of the contaminant fraction of DNA obtained from an ancient giant panda bone. *BMC Research Notes*, 10(1), 754. <https://doi.org/10.1186/s13104-017-3061-3>.
- Dabney, J., Knapp, M., Glock, I., Gansauge, M., Weihmann, A., Nickel, B., Valdiosera, C., García, N., Pääbo, S., Arsuag, J. & Meyer, M. (2013). Complete mitochondrial genome sequence of a Middle Pleistocene cave bear reconstructed from ultrashort DNA fragments. *PNAS*, 110(39), 15758-15763.
- Rohland N, Siedel H, Hofreiter M. (2004). Nondestructive DNA extraction method for mitochondrial DNA analyses of museum specimens. *Biotechniques*, 36(5), 814-821.
- Rohland N, Hofreiter M (2007) Comparison and optimization of ancient DNA extraction. *Biotechniques*, 42(3):343–352.
- Rohland, N., Hofreiter, M. (2007). Ancient DNA extraction from bones and teeth. *Nature Protocols*, 2, 1756–1762 <https://doi.org/10.1038/nprot.2007.247>
- Rohland, N., Siedel, H. & Hofreiter, M. (2010). A rapid column-based ancient DNA extraction method for increased sample throughput. *Molecular Ecology Resources*, 10, 677–683.
- Rohland N, Glocke I, Aximu-Petri A, Meyer M. (2018). Extraction of highly degraded DNA from ancient bones, teeth and sediments for high-throughput sequencing. *Nature Protocols* 13(11), 2447-2461. doi: 10.1038/s41596-018-0050-5.
- Sambrook J, Russell D (2001) Molecular Cloning: A Laboratory Manual. USA, New York: Cold Spring Harbor Laboratory Press.

2 LABORATORY EQUIPMENTS

- pipettes (range 1 – 5000 µl)
- vortex mixer
- Qubit® Fluorometer
- Thermocycler
- Precision balance
- 15600 g suitable centrifuge for 1.5 and 2 ml tubes
- >940 g suitable centrifuge suitable for 50 ml Falcon tubes

- UV cabinet
- laboratory oven
- rotator (2 ml tubes)
- fume hood
- DNA contamination removal solution
- Racks for microcentrifuge and falcon tubes
- Glassware for solution preparation
- Parafilm, tape and laboratory marker pen

3 PLASTIC WARE

Product	Company	Catalogue number
10-5000 µl tips (with filter)	STARLAB	S1121-2710; S1120-3810; I1011-5830; S1122-1830; I1009-5000
Eppendorf Tube 1.5 and 2 ml	EPPENDORF	0030108051; 0030108078
Zymo Columns	ZYMO	C1016-50
Weighing Boats	A.HARTENSTEIN	W100
Single use scalpels	A.HARTENSTEIN	SK20
1.5 ml Low Retention Tube	THERMO FISHER	3451
Falcon Tube 15 and 50 ml	STARLAB	E1450-0200; E1415-0200

4 REAGENTS

Reagent	Company	Catalogue number
Guanidine thiocyanate*	ROTH	0017.3
Proteinase K#	PROMEGA	V3021
NaCl 5M	THERMO FISHER	AM9760G
Tris-HCl 1M pH8	THERMO FISHER	15568-025
EDTA	VWR	E177-500MLDB
Tween-20 100%	A.HARTENSTEIN	CT21
2-Mercaptoethanol	SIGMA ALDRICH	M6250-10ML
HPLC water	A.HARTENSTEIN	CW20
Guanidine hydrochloride	ROTH	0037.1
Isopropanol	A.HARTENSTEIN	CP50
Sodium Acetate 3M	VWR	J61928.AE
MinElute Kit (also containing PE buffer)	QIAGEN	28006

* Only for guanidine treatment extraction # Only for Proteinase K treatment

5 PREPARATION OF LABORATORY WORK

DNA extraction should be carried out in dedicated archival DNA laboratories. Clean your workspace with DNA contamination removal solution and UV light before starting each extraction experiment. Information on how to set up a dedicated lab can be found here: Fulton, T.L., & Shapiro B. (2019) Setting Up an Ancient DNA Laboratory. In Shapiro, B., Barlow, A., Heintzman, P., Hofreiter, M., Paijmans, J., Soares, A. (Eds.) *Ancient DNA: Methods in Molecular Biology* (pp.1-13). USA, New York: Humana Press.
https://doi.org/10.1007/978-1-4939-9176-1_1

6 PREPARATION OF BUFFERS

Depending on the chosen treatment, prepare the corresponding extraction buffer.

IMPORTANT NOTE: the buffers contain 2-mercaptoethanol. Archival DNA Extraction protocol steps 1 to 5 should be performed in a fume hood and corresponding safety measures should be met. The 2-mercaptoethanol and Guanidinium thiocyanate buffer waste should be discarded appropriately.

6.1 Guanidine Treatment

Guanidinium thiocyanate buffer is adapted from Rohland et al. (2004) but the N-phenacylthiazolium bromide (PTB) is replaced with 2-mercaptoethanol and Triton X100 with Tween 20. Make up fresh each time.

Extraction buffer	Final concentration
GuSCN powder pure	5 M
5 M NaCl	25 mM
1M Tris, pH 8.0	50 mM
0.5 M EDTA, pH 8.0	20 mM
10 % Tween-20	1 %
2-mercaptoethanol	1 %
ddH ₂ O	-

6.2 All Proteinase K treatment extraction buffers:

Make up fresh each time. UV- irradiate the solution before adding proteinase K.

Extraction buffer	Final concentration
5 M NaCl	100 mM
1M Tris-Cl	10 mM
0.5 M EDTA, pH 8.0	25 mM
20 % SDS	0.5 %
Proteinase K 10mg/ml	0.1 mg/ml
ddH ₂ O	-

6.3 Binding buffer

Can be stored for 1 month. UV- radiate the solution before use.

Binding buffer	Final concentration
Guanidine hydrochloride	5 M
Isopropanol	40 %
10% Tween20	0.05 %
Sodium acetate 3M solution	90 mM
ddH ₂ O	-

6.4 TET elution buffer

Can be stored indefinitely. UV- irradiate the solution before use.

TET buffer	Final concentration
1M Tris-HCL solution, pH 8.0	10 mM
0.5M EDTA solution	1 mM
10% Tween20 solution	0,05 %
ddH ₂ O	-

7 PRE-EXTRACTION PROCEDURES

- Label all tubes necessary for extraction
- Fill 15 ml falcon tubes with 13 ml binding buffer

7.1 Prepare binding apparatus

Material:

- Zymo spin V column extension reservoir
- Qiagen MinElute silica spin column
- 50 ml falcon tube

Methods:

7.1.1 Remove the MinElute column from the collection tube and cut off the lid of the MinElute column., Keep then lids for later. The lid can be pushed into the top of the collection tube to maintain sterility.

7.1.2 Assemble the binding apparatus by fitting the MinElute column onto the extension reservoir. Label the side of the MinElute column.

7.1.3 Put the reservoir with MinElute column inside the 50 ml Falcon tube.

Note: the falcon tube lid no longer fits properly and will therefore be taped to the tube in step 8.3.

7.2 Sample preparation before extraction

7.2.1 Start with weighing the sample; we recommend starting material to be up to 60 mg at maximum.

7.2.2 Cut the sample into pieces using a sterile scalpel and place in a 2 ml low retention microcentrifuge tube.

7.2.3 Add 1 ml PE buffer to the sample and pipette up and down.

7.2.4 Pipette off and discard PE buffer

7.2.5 Repeat steps 7.2.3 and 7.2.4

7.2.6 Proceed to the next step immediately.

Optional: heat sample at 95°C for 15 minutes in 1 ml proteinase K extraction buffer.

Note: if heating the sample, the extraction buffer should not yet contain proteinase K! Add proteinase K after heating. Also see details on the 95 °C prior Proteinase K treatment.

8 ARCHIVAL DNA EXTRACTION

8.1 Add 1ml extraction buffer depending on the treatment applied to the sample and incubate at corresponding temperature (37° C or 65° C) with constant rotation for ~18 hours. Seal lids of 2 ml tubes with parafilm to prevent leakage.

8.2 Spin down the tubes to collect tissue remains at the bottom. Depending on the type of tissue, this may need spinning at 15,600 g for 2 minutes.

8.3 Pipette off supernatant and add to 15 ml falcon containing 13 ml binding buffer. Invert manually 2 – 3 times. Then pour binding buffer and sample into the binding apparatus reservoir (with labelled spin column + lid). Tape the lids onto the binding apparatus.

8.4 Centrifuge the binding apparatus for 4 minutes at 940 g, then rotate the binding apparatus 90° and centrifuge again at 940 g for another 4 minutes.

Optional: repeat centrifugation if there is still liquid in the binding apparatus reservoir.

8.5 Detach the MinElute column from the binding apparatus and put in the original 2 ml collection tube and replace the original cap, then dry spin for 1 minute at 3,300 g. In case there is still liquid in the MinElute column, an increase in centrifugation speed may be necessary.

8.6 Wash the silica membrane by adding 650 µl PE buffer, centrifuging at 900 g for 1 minute. Discard the flow through.

8.7 Repeat step 8.6.

8.8 Dry spin the MinElute column at 15,600 g for 1 minute and place in a clean low retention 1.5 ml tube, labeled accordingly.

8.9 DNA elution is performed in two steps. First, pipette 12.5 µl TET buffer directly to the silica membrane, incubate for 10 minutes, and centrifuge at 15,600 g for 30 seconds.

8.10 Repeat step 8.9 to attain a total volume of 25 µl DNA extract. Measure the DNA content with fluorometer (e.g. Qubit high sensitivity dsDNA kit)

Optional: when performing the “Proteinase K re-extraction protocol, use the tissue remains from step 8.2, and re-start the protocol from step 8.1 replacing the GuSCN buffer with the Proteinase K extraction buffer.

Supplemental Material 2

SINGLE STRANDED LIBRARY PREPARATION PROTOCOL

1. ESSENTIAL READING

This protocol is based on and modified from the references listed below. If using this protocol, please read and cite:

Gansauge M-T, & Meyer M. (2013). Single-stranded DNA library preparation for the sequencing of ancient or damaged DNA. *Nature Protocols*, 8, 737–748.

Meyer, M., Kircher, M., Gansauge, M. T., Li, H., Racimo, F., Mallick, S., ... Pääbo, S. (2012). A high-coverage genome sequence from an archaic Denisovan individual. *Science*, 338(6104), 222–226. <https://doi.org/10.1126/science.1224344>

Korlević P, Gerber T, Gansauge M-T, Hajdinjak M, Nagel S, Aximu-Petri A, Meyer M. (2014). Reducing microbial and human contamination in DNA extractions from ancient bones and teeth. *Biotechniques*, 59, 87–93.

2. LABORATORY EQUIPMENT

For a successful conduction of the protocol, the laboratory should be equipped with the following:

- standard pipettes (range 1 – 1000 µl)
- vortex mixer
- table centrifuge for 0.2 ml and 1.5 ml tubes
- thermomixer
- thermocycler
- scale for preparing solutions
- UV cabinet
- rotator (0.2 ml and 1.5 ml tubes)
- DNA contaminate removal solution
- 1.5 ml and 0.2 ml magnetic racks
- 1.5 ml and 0.2 ml cooling racks
- Glassware for solution preparation
- Racks for microcentrifuge
- Laboratory marker pen

3. PLASTIC WARE

Product	Company	Catalogue number
10-1000 µl tips (with filter)	STARLAB	S1121-2710; S1120-3810; I1011-5830; S1122-1830

Eppendorf Tube 1.5 and 2 ml	EPPENDORF	0030108051; 0030108078
1.5 ml Low Retention Tube	THERMO FISHER	3451
8 strip tubes with lids attached	STARLAB	I1402-3700

4. REAGENTS

Reagent	Company	Catalogue number
SDS 20%	THERMO FISHER	AM9820
NaCl 5M	THERMO FISHER	AM9760G
Tris-HCl 1M pH8	THERMO FISHER	15568-025
EDTA	VWR	E177-500MLDB
Tween-20 100%	ROTH	CT21
SSC Buffer 20X	SIGMA ALDRICH	1054.1
HPLC water	A.HARTENSTEIN	CW20
Circligase II	BIOZYM	131402(CL9021K)
Endonuclease VIII	NEB	M0299S
Afu UDG	NEB	M0279S
Fast AP	THERMO FISHER	EF0651
Dynabeads My One Streptavidin	THERMO FISHER	C1-65001
PEG-4000 solution, 50%	SIGMA-ALDRICH	89782-100ML-F
DNA Polymerase I (Large)	NEB	M0212S
Klenow fragment		
T4 DNA Ligase	THERMO FISHER	EL0011
MinElute Kit	QIAGEN	28006

5. CUSTOM OLIGOS (Gansauge & Meyer 2013)

Oligo name	Sequence
CL78	[Phosphate]AGATCGGAAG[C3Spacer] ₁₀ [TEG-biotin]
CL9	GTGACTGGAGTTCAGACGTGTGCTCTTCCGATC
CL104	[Phosphate]TCGTCGTTTGGTATGGCTTCATTCAGCTCCGGTTCCTCAACG ATCAAGGCGAGTTACATGA[Phosphate]
CL53	CGACGCTCTTC-[ddc]
CL73	[Phosphate]GGAAGAGCGTCGTAGGGAAAGAG*T*G*T*A

TEG = triethylene glycol spacer; * = phosphothioate linkage; Ddc = dideoxycytidine

6. PREPARATION OF LABORATORY WORK

- Single stranded library preparation should be carried out in dedicated archival DNA

laboratories. Information on how to set up a dedicated lab can be found here: Fulton, T.L., & Shapiro B. (2019) Setting Up an Ancient DNA Laboratory. In: Shapiro, B., Barlow, A., Heintzman, P., Hofreiter, M., Paijmans, J., Soares, A. (eds) Ancient DNA. Methods in Molecular Biology, vol 1963 (pp.1-13). Humana Press, New York, NY. https://doi.org/10.1007/978-1-4939-9176-1_1

- Clean your workspace with DNA-Exitus or similar.
- Label all tubes necessary for library preparation accordingly.
- The use of uracil-DNA glycosylase (UDG) is optional.
- You can optionally generate a library using 1µl of 0.1µM CL104 synthetic oligonucleotide as a positive control in a separate reaction. This allows the efficiency of conversion of DNA fragments into library to be measured using qPCR.
- Include a library blank, i.e. a reaction blank without sample DNA or oligonucleotides.

7. PREPARATION OF BUFFERS

7.1 Bead-binding buffer (10 ml)

Prepare fresh for each library preparation. After addition of 20% SDS use immediately.

- 7.58 ml HPLC grade water
- 2 ml of 5M NaCl
- 100 µl of 1M Tris-HCl (pH 8.0)
- 20 µl of 0.5 M EDTA (pH 8.0)
- 50 µl of 10% Tween
- 250 µl of 20% SDS

TIP: prepare after step 8.1 during library preparation.

7.2 Wash buffer A (50 ml)

Storable at room temperature for 1 month

- 46.9 ml HPLC grade water
- 1 ml of 5 M NaCl
- 500 µl of 1 M Tris-HCl (pH 8.0)
- 100 µl of 0.5 M EDTA (pH 8.0)
- 250 µl of 10% Tween
- 1.25 ml of 20% SDS

7.3 Wash buffer B (50 ml)

Storable at room temperature for 1 year

- 48.15 ml HPLC grade water
- 1 ml of 5 M NaCl
- 500 µl of 1 M Tris-HCl (pH 8.0)

- 100 µl of 0.5 M EDTA (pH 8.0)
- 250 µl of 10% Tween 20

7.4 Stringency wash buffer (50 ml)

Storable at room temperature for 1 month

- 49.5 ml of HPLC grade water
- 250 µl of 20% SDS
- 250 µl of 20× SSC buffer

7.5 Stop solution (100 µl)

Storable at room temperature for 1 year

- 98 µl of 0.5 M EDTA (pH 8.0)
- 2 µl of Tween 20 (undiluted)

7.6 TE buffer (50 ml)

Storable at room temperature for 1 year

- 49.4 ml of HPLC grade water
- 500 µl of 1 M Tris-HCl (pH 8.0)
- 100 µl of 0.5 M EDTA (pH 8.0)

7.7 TET buffer (50 ml)

Storable at room temperature for 1 year

- 49.15 ml of HPLC grade water
- 500 µl of 1 M Tris-HCl
- 100 µl of 0.5 M EDTA
- 250 µl of 10% Tween

7.8 Double-stranded adapter (100 µM)

- Set up the following reaction:

Reagent	µl
TE buffer	9.5
NaCl (5M)	0.5
CL53 (500 µM)	20
CL73 (500 µM)	20

- Incubate the 50 µl reaction in a thermal cycler for:
 - 10 s at 95 °C
 - decrease temperature by 0.1 °C per second, until reaching 14 °C
- Then add 50 µl of TE buffer to the hybridized adapter to obtain a concentration of 100 µM in 100 µl. The resulting solution can be stored at –20 °C for at least 1 year

TIP: buffers 7.2. to 7.7 should be UV radiated for 4 hours before usage. Optional: buffer 7.1 can be UV radiated prior to adding 20% SDS.

8. SINGLE STRANDED LIBRARY PREPARATION

DAY 1

8.1 Uracil excision and DNA cleavage at abasic sites

8.1.1 Make up the following reaction mix:

Reagent	FC in reaction	µl per sample
Nuclease free water	-	11
Circligase buffer II (10x)	1.8x (1x in stage 3)	8
MnCl	4.5 mM (2.5 mM in stage 3)	4
Endonuclease VIII (10 U/µl)	0.11 U/µl	0.5
Afu UDG (2 U/µl)	0.02 U/µl	0.5

8.1.2 Aliquot 24 µl into 0.2 ml PCR strip tubes.

8.1.3 Add 20 µl DNA extract or a recommended maximum of 13 ng DNA to each (or nuclease free water for library blank). The total reaction volume is now 44 µl.

8.1.4 Mix gently and spin down, incubate in a thermal cycler for 1 hour at 37 °C.

TIP1: For the CL104 positive control add 1 µl of 0.1 µM CL104 oligo and 19 µl TET buffer.

TIP2: prepare buffer 7.1 during the waiting time.

8.2 Dephosphorylation and heat denaturation

8.2.1 Add 1.0 µl of FastAP (1U) to each 0.2 ml tube from 8.1.4, mix gently and spin.

8.2.2 Incubate in a thermal cycler for 10 min at 37°C and 2 min at 95°C.

8.2.3 While the thermal cycler is still at 95 °C, quickly transfer the 0.2 ml tubes into a freezer cooling block. Cool for at least 1 min, spin down gently in a table centrifuge and return to room temperature.

8.3 Ligation of the first adapter

8.3.1 Add the following to each 0.2 ml tube from step 8.2: 32 µl PEG-4000 (50%), 1 µl adapter oligo CL78 (10 µM) - Final volume is 80.0 µl.

8.3.2 Vortex and spin. Add 2.0 µl of Circligase II to each 0.2 ml tube, mix gently by pipetting up and down.

8.3.3 Incubate in a thermal cycler overnight at 60 °C.

8.3.4 Add 2 µl of stop solution to each 0.2 ml tube from step 8.3.3, vortex and spin down.

TIP: Ligation products can be stored at – 20 °C for several days

DAY 2

8.4 Immobilization of ligation products on beads

- 8.4.1 Resuspend Dynabeads by vortexing.
- 8.4.2 Pipette and combine 20 µl of bead suspension for each sample into a 1.5 ml tube.
- 8.4.3 Pellet the beads using a magnetic rack, discard supernatant, wash with 500 µl of bead-binding buffer (see 5.1).
- 8.4.4 Repeat step 8.4.3.
- 8.4.5 Dry the beads from step 8.4.5 and resuspend the dry beads in 250 µl bead-binding buffer.

TIP: When using more than 5 samples at a time, spread out the bead washing steps to a corresponding number of 1.5 ml tubes.

- 8.4.6 For each sample, transfer 250 µl of bead suspension to a 1.5 ml low retention tube.
- 8.4.7 Incubate the ligation reactions from step 8.3.4 in a thermal cycler for 1 min at 95 °C.
- 8.4.8 While still at 95°C, quickly transfer tubes into a 0.2 ml tube cooling block and cool for 1 min. Return to room temperature and spin down briefly.
- 8.4.9 Add the ligation reactions to the bead suspensions prepared in step 8.4.6.
- 8.4.10 Rotate for 20 min at room temperature.
- 8.4.11 Pellet the beads in a 1.5 ml tube magnetic rack. Discard supernatants, remove tubes from magnetic rack and add 200 µl wash buffer A (see 7.2). Pipette up and down and pellet beads again in the magnetic rack. Remove supernatant
- 8.4.12 Repeat step 8.4.11 **but replace wash buffer A with wash buffer B** (see 7.3.). Pellet the beads and move to step 8.5 immediately.

8.5 Primer annealing and extension

- 8.5.1 Pre-heat the thermomixer to 25°C.
Prepare the following master mix (48 µl per sample):

Reagent	FC in 50 µl	µl per sample
Nuclease free water	-	41.6
Klenow buffer (10x)	1x	5
dNTP mix (25 mM each)	200 µM	0.4
Extension primer CL9 (100 µM)	2 µM	1

- 8.5.2 Add 48 µl master mix, resuspend beads by vortexing.
- 8.5.3 Preheat thermal cycler to 65 °C, incubate 1.5 ml tubes for 2 min.
- 8.5.4 Transfer tubes to a cooling block for 1 min, then immediately to the thermomixer preheated to 25 °C.
- 8.5.5 While the tubes are in the thermomixer, add 2 µl of Klenow Fragment of DNA Polymerase 1 (10 U/µl) to each tube.
- 8.5.6 Incubate the reactions in the thermomixer for 5 min at 25°C and then 25 min at 35°C. The solution needs mixing every minute.
- 8.5.7 Once step 8.5.6 is finished, remove tubes from the thermo mixer and preheat the Thermomixer to 45 °C. Pellet beads, discard supernatant.
- 8.5.8 Wash beads by adding 200 µl wash buffer A. Pipette up and down and pellet the beads. Resuspend in 100 µl stringency wash buffer (see 7.4).
- 8.5.9 Incubate for 3 min at 45°C in the thermomixer, mixing every 30 secs.
- 8.5.10 Pellet beads, discard supernatant, wash in 200 µl wash buffer B. Pellet the beads and move to step 8.6 immediately.

TIP: Washing steps are as in 8.4.11/12 but with an additional washing step with stringency buffer between washing with buffers A and B.

8.6 Ligation of second adapter and library elution

- 8.6.1 Pre-heat the thermomixer to 22°C now and a thermal cycler to 95°C.
- 8.6.2 Prepare the following reaction mix (98 µl per sample):

Reagent	FC in 100 µl	µl per sample
Nuclease free water	-	73.5
T4 DNA ligase buffer (10x)	1x	10
PEG-4000 (50%)	5%	10
Tween 20 (1% solution)	0.025%	2.5
Double stranded adapter (100 µM)	2 µM	2

- 8.6.3 Add 98 µl reaction mix to the pellets from 8.5.10 and resuspend by vortexing
- 8.6.4 Add 2 µl of T4 DNA ligase (10 U), vortex, then incubate in a thermomixer for 1 h at 22°C. The solution needs mixing every minute.
- 8.6.5 Once step 8.6.4 is finished, remove tubes from the thermomixer and preheat the thermomixer to 45 °C. Pellet beads, discard supernatant.
- 8.6.6 Wash beads by adding 200 µl wash buffer A. Pipette up and down and pellet the beads. Resuspend in 100 µl stringency wash buffer (see 7.4).
- 8.6.7 Incubate for 3 min at 45°C in the thermomixer, mixing every 30 secs.
- 8.6.8 Pellet beads, discard supernatant, wash in 200 µl wash buffer B. Pellet the beads.
- 8.6.9 Add 25µL of TET buffer, remove from magnetic rack and resuspend the beads by pipetting.

- 8.6.10 Transfer the bead suspension to 0.2 ml tubes, spin down briefly.
- 8.6.11 Incubate the 0.2 ml tubes in a thermocycler for 1 min at 95 °C and thereafter immediately transfer the PCR strip tubes to a 96-well magnetic rack.
- 8.6.12 Transfer the supernatant, which contains the library molecules, to new, pre-labeled low retention 0.2 ml tubes. Discard the beads.

TIP: Single stranded libraries can be stored at – 20 °C for several months.

The libraries can now be amplified and indexed for sequencing on an Illumina® short read sequencer, for example a NextSeq500/550 sequencer. Note that custom primers cannot be used on all Illumina systems. An amplification protocol is given in Gansauge & Meyer (2013).

Supplementary Material 3

REFERENCES USED FOR FASTQSCREEN ANALYSES

1. Fixed set of references checked for all samples:

Adapters, Vectors, rRNA, *Escherichia coli* genome & *Homo sapiens* genome:

Provided packages for download in Fastqscreen:

https://www.bioinformatics.babraham.ac.uk/projects/fastq_screen/_build/html/index.html#installation

Wingett, S. W., & Andrews, S. (2018). FastQ Screen: A tool for multi-genome mapping and quality control. *F1000Research*, 7, 1338. doi:10.12688/f1000research.15931.2

***Anolis carolinensis* genome:**

ftp://ftp.ensembl.org/pub/release-98/fasta/anolis_carolinensis/dna/

Alföldi, J., Di Palma, F., Grabherr, M., Williams, C., Kong, L., Mauceli, E., ... Lindblad-Toh, K. (2011). The genome of the green anole lizard and a comparative analysis with birds and mammals. *Nature*, 477(7366), 587–591. doi: 10.1038/nature10390.

***Danio rerio* genome:**

http://www.ensembl.org/Danio_rerio/Info/Index

Howe, K., Clark, M.D., Torroja, C. F., Torrance, J., Berthelot, C., Muffato, M., ... Stemple, D.L. (2013). The zebrafish reference genome sequence and its relationship to the human genome. *Nature*, 496, 498. Available at: <https://doi.org/10.1038/nature12111>.

***Xenopus laevis* genome:**

http://www.xenbase.org/common/displayJBrowse.do?data=data/xl9_2

Karimi, K., Fortriede, J.D., Lotay, V.S., Burns, K.A., Wang, D.Z., Fisher, M.E., ... Vize, P.D. (2018). Xenbase: a genomic, epigenomic and transcriptomic model organism database. *Nucleic acids research*, 46, 861–868. doi: 10.1093/nar/gkx936.

***Rhincodon typus* genome:**

<https://www.ncbi.nlm.nih.gov/nuccore/LVEK000000000.1>

Read, T. D., Petit III, R.A., Sandeep, J.J., Alaam, Md.T., Weil, M.R., Ahmad, M., ... Dove, A.D.M. (2017). Draft sequencing and assembly of the genome of the world's largest fish, the whale shark: *Rhincodon typus* Smith 1828. *BMC Genomics*, 18(1), 532. doi: 10.1186/s12864-017-3926-9.

2. Specific references used for evaluating relative endogenous DNA content:

***Etmopterus spinax* transcriptome:**

<https://www.ncbi.nlm.nih.gov/nuccore/GHAY000000000>

Delroisse, Duchatelet, L., Flammang, P., & Mallefet, J. (2019). De novo transcriptome analyses provide insights into opsin-based photoreception in the lanternshark *Etmopterus spinax*. *PLOS ONE*, 13(12), E0209767. <https://doi.org/10.1371/journal.pone.0209767>.

***Carcharodon carcharias* transcriptome:**

<https://bmcbgenomics.biomedcentral.com/articles/10.1186/1471-2164-14-697#Sec14>

Richards, V. P., Suzuki, H., Stanhope, M. J., & Shivji M. S. (2013). Characterization of the heart transcriptome of the white shark (*Carcharodon carcharias*). *BMC Genomics* 14, 697.

doi:10.1186/1471-2164-14-697

***Scyliorhinus torazame*, *Chiloscyllium punctatum* genomes:**

ddbj.nig.ac.jp/DRAsearch/submission?acc=DRA006338

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NCBI Sequence Read Archive accession number: BioProject PRJNA625657, BioSample SAMN14604402

Supplementary Table S2: Long-range PCR primer pairs used for bait-library preparation

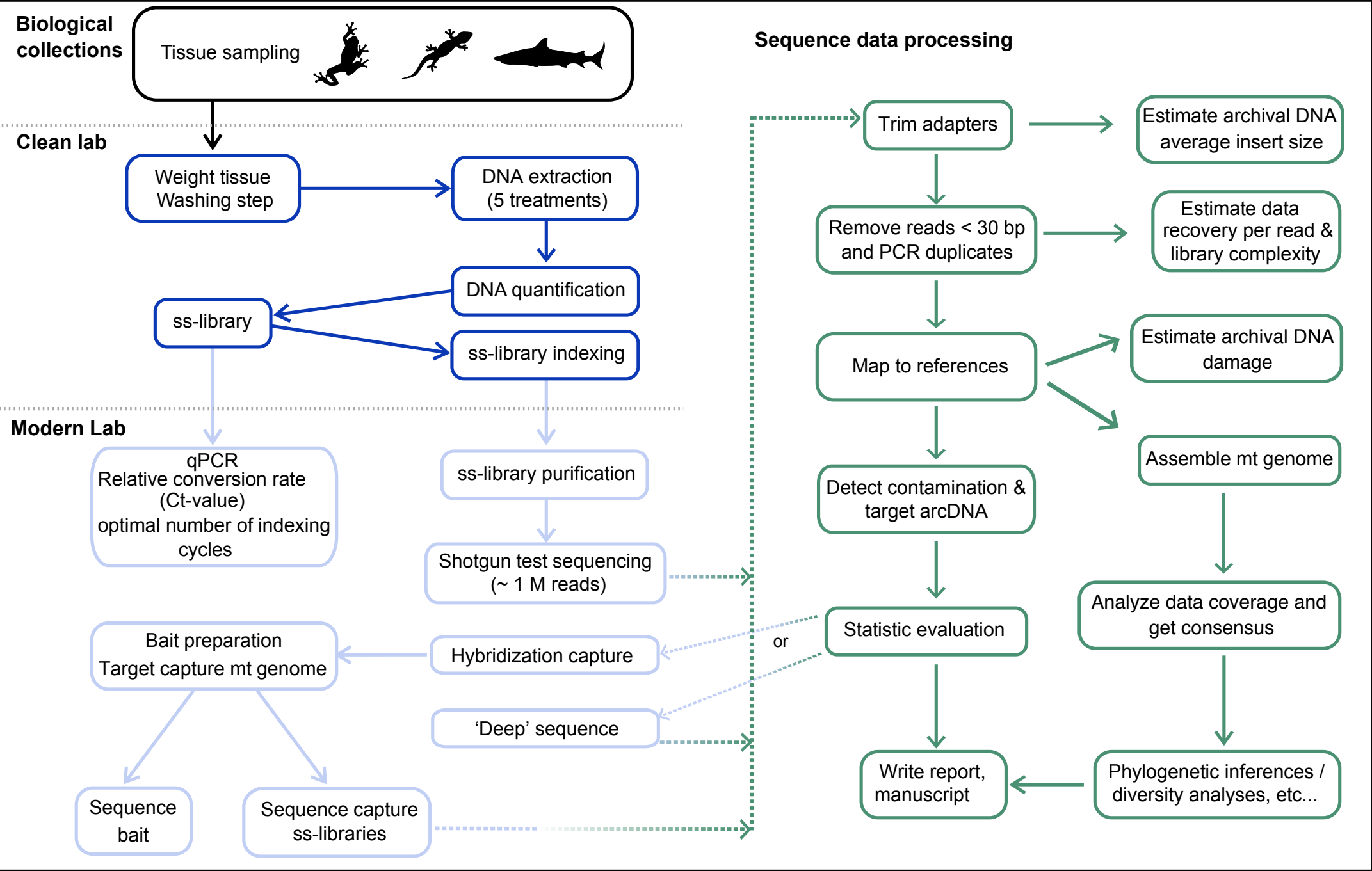
Species selected for baits libraries	Primer name	Sequence (5'→3')	Reference
<u>Lissamphibia</u>			
<i>Allobates olfersioides</i> ,	12SAL	AAACTGGGATTAGATACCCCACTAT	Kessinger et al 1989
<i>Aplastodiscus arildae</i> ,	FCOIII9400_H	AGCCTCATCARTAAATTGAGAC	This Work
<i>Paratelmatobius cardosoi</i> , <i>Thoropa saxatilis</i>	FCOIIIBeg_L	CAAGAAAACACYTAATGGC	This Work
	FCB14400Hmod	CAGATAAAGAARAATGADGCKCC	Modified from Zhang et al 2013
<u>Chondrichthyes</u>			
<i>Etmopterus molleri</i>	Emops_lr_F1,	GAGGTCTGGGACACAAAGAACTGCTAATTC	this study
	Emops_lr_R1	GACAAATGATTACGCTACCTTCGCACGGTC	this study
	Emops_lr_F2	CTCGGCAAACATAAACTCCGCCTGTTTACCAAAAAC	this study
	Emops_lr_R2	TTGCACCAAGAGTTTTTGGTTCCTAAGACC	this study
<i>Leucoraja erinacea</i>	Leuco_lr_F1_2379	CAAGGAGTAAAATAAAACCTCCTTATCGAT	this study
	Leuco_lr_R1_12220	GATGGAGTAGAGGTCAAATTTAAAGCTTAT	this study
	Leuco_11194_lr_F2	AACAGATCTTAAATCCTTAATTGCCTACTC	this study
	Leuco_3297_lr_R2	ATTAAGCCTAGGGTAACTTCATAGGAAATA	this study

References:

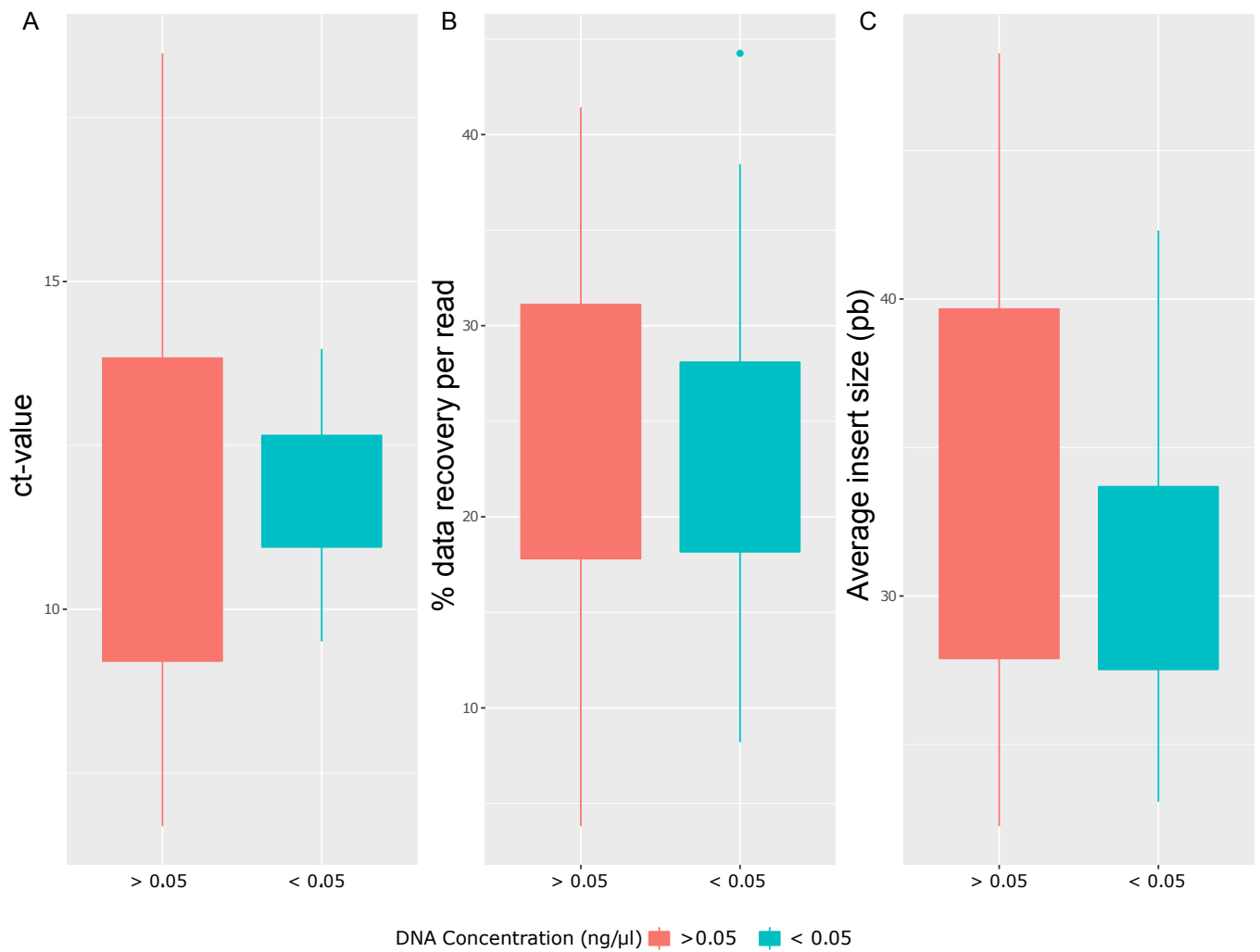
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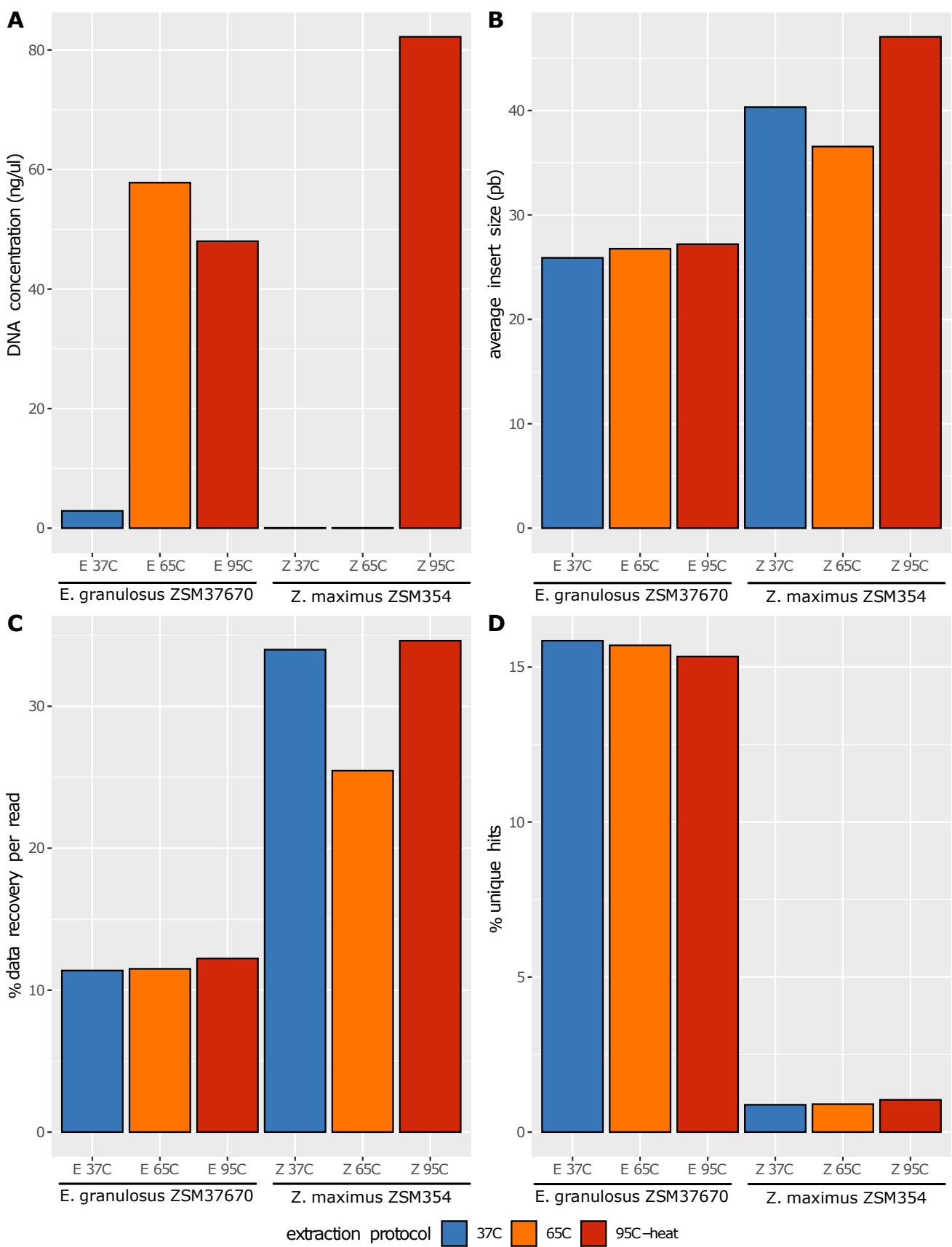
Supplementary Figure S1: Graphical summary of laboratory and data analysis steps.



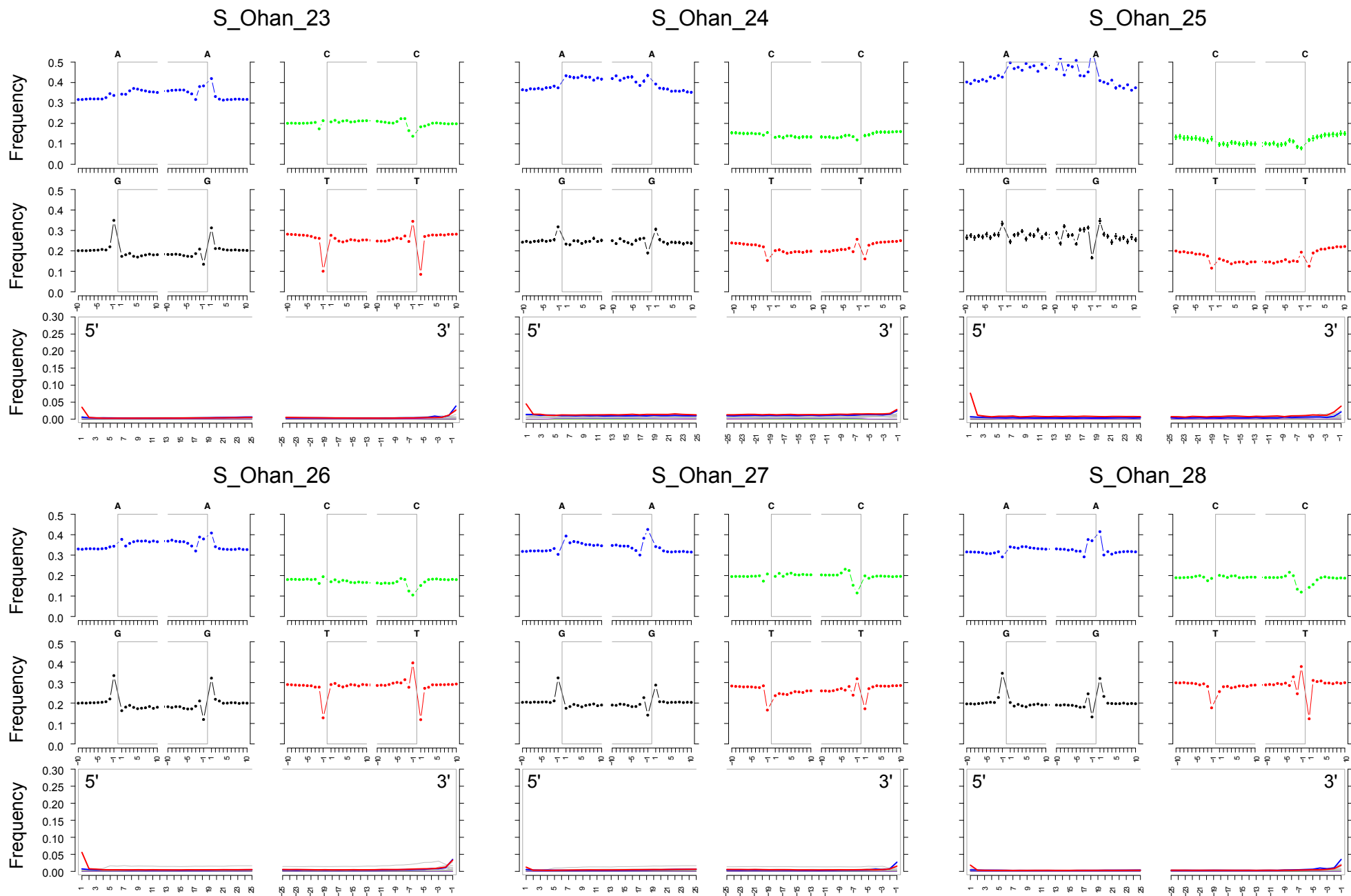
Supplementary Figure S2: Comparison of libraries constructed from DNA extracts where no DNA was detected (< 0.05 ng/ μ l) and libraries prepared from extracts with detectable DNA amounts.



Supplementary Figure S3: Comparisons of variables collected during extraction and library preparation of two samples extracted under different conditions (Proteinase K treatment, Proteinase K 65 °C treatment and 95 °C prior Proteinase K treatment)



Supplementary Figure S4: Damage patterns estimated with MapDamage2 of king cobra samples (*Ophiophagus hannah*). Upper four plots show base frequencies of reads (area in brackets) and surrounding regions (outside brackets). Red: C to T substitutions; blue: G to A substitutions; grey: all other substitutions.



Supplementary Figure S5: Adenine and Guanine ratios in comparison to the total number of bases detected at position -1 of the 5' end. Information collected from mitochondrial archival DNA of target captured (N=5) and King cobra samples (N=6).

